Getting Started with Xamarin.Forms in Visual Studio 2017

Compiled by Thayne Breetzke

1. Install or Set up Visual Studio

Note: The system requirements for Visual Studio 2017 can be found here.

a) If you **do not** have Visual Studio 2017 installed, follow the guide below to install it with *Xamarins.Forms* support:
https://docs.microsoft.com/en-za/xamarin/cross-platform/get-started/installation/index?tabs=windows

The steps are summarized as follows:

1. Run the Visual Studio 2017 installer.
2. Select the **Mobile development with .NET** workload from the installation screen.
3. You will probably also need the following workloads, depending on your needs, so select them as well:
   a. **.NET desktop development**
   b. **Universal Windows Platform development**
   c. **ASP.NET and web development**
   d. **Data storage and processing**
4. Review the **Installation details** panel on the right (for each selected workload).
5. Click the **Install** button.
6. Once the installation has completed, click the **Launch** button to start Visual Studio.

b) If you **already have** Visual Studio 2017 installed, check whether it has *Xamarin* installed by following the guide below:
https://docs.microsoft.com/en-za/xamarin/cross-platform/get-started/installation/windows#adding-xamarin-to-visual-studio-2017

The steps are summarized as follows:

1. Start Visual Studio 2017.
2. Click the **Help** menu.
3. If Xamarin is installed, you should see a Xamarin menu item.

If it is installed, proceed to *Step 2* below, otherwise continue with the steps below to add the **Mobile development with .NET** workload.
https://docs.microsoft.com/en-za/visualstudio/install/modify-visual-studio?view=vs-2017

The steps are summarized as follows:

1. Run the Visual Studio 2017 installer in your **Start** menu.
2. Click the **Modify** button.
3. Select the **Mobile development with .NET** workload (and any other workloads you need – see above).
4. Click the **Modify** button again.
5. After the new workloads and components are installed, click the **Launch** button.

2. Install the latest Android components (may already be included in VS 2017)

You should have the latest Android SDK Tools and Android API platform installed. You can update to the latest versions using the Android SDK Manager.

The steps are summarized as follows:

1. Start Visual Studio 2017 as Administrator (Start > right-click *Visual Studio 2017* > More > Run as administrator).
2. Go to Tools > Android > Android SDK Manager.
3. If the Android SDK Location box is empty, click the … and browse to the *C:\Program Files (x86)\Android\android-sdk* folder
4. If no SDK component information is listed, you may need to click the settings icon in the bottom-right corner of the window and select Repository > Google
5. Click the **Updates Available** button to install all the available updates.
6. You should also select the latest API Platform and SDK Tools and click **Apply Changes** to install them.

3. Install the Java SDK (may already be included in VS 2017)

Follow these steps to install the latest supported Java SDK:

1. Follow the link below to install the Java SDK. Install the latest SDK (do not install JDK 9 or newer as it is not supported – install the latest JDK 8).
   http://www.oracle.com/technetwork/java/javase

   The latest update is available here:
   http://www.oracle.com/technetwork/java/javase/downloads/jdk8-downloads-2133151.html

2. Restart your computer.

Note that *Xamarin.Android* does not support JDK 9.

4. Enable Intel hardware acceleration

Follow these steps to allow your apps to deploy to the Android emulator:

1. Enable Intel Virtualization in your computer's BIOS settings by rebooting and entering the BIOS settings). You will have to press/hold a certain key (e.g. ESC or DEL) when the computer starts so that you can enter the BIOS settings menu.

2. Download and extract Intel's Hardware Accelerated Execution Manager (HAXM) 7.4.1 or latest:
   https://github.com/intel/haxm/releases/tag/v7.4.1 [haxm-windows_v7_4_1.zip]

3. Run the **haxm_check.exe** program to check whether your hardware supports HAXM.

4. Run the **intelhaxm_android.exe** program to install the HAXM software.

More information is available here:
https://docs.microsoft.com/en-za/xamarin/android/get-started/installation/android-emulator/hardware-acceleration?pivots=windows

5. View the Microsoft Video: *Intro to Xamarin.Forms for Visual Studio 2017*

This video introduces you to *Xamarin.Forms* for Visual Studio 2017.

https://channel9.msdn.com/Events/Xamarin/Xamarin-University-Presents-Webinar-Series/Intro-to-XamarinForms-for-Visual-Studio-2017

6. Do the Xamarin tutorial: *Getting Started with Xamarin.Forms*

Follow the link below:
https://developer.xamarin.com/guides/xamarin-forms/getting-started/

The sections are listed below:

1. Introduction
2. Requirements
3. Build your first app
4. Hello, Xamarin.Forms
    a. Part 1: Quickstart
    b. Part 2: Deep Dive
5. Hello, Xamarin.Forms Multiscreen
    a. Part 1: Quickstart
    b. Part 2: Deep Dive
6. Introduction to Xamarin.Forms

Note that when you create a new Xamarin app, you will need to update the NuGet packages as follows:

1. Right-click on the solution (not on any of the projects).
2. Select **Manage NuGet packages...**
3. Install the updates.

Note that the target/compile version for Android projects must be set to *Use latest installed platform*. However, the minimum version can be set to API 19 so you can continue to support devices that use Android 4.4 and newer. These values are set in the **Project Options**. See below for more information:
https://docs.microsoft.com/en-za/xamarin/xamarin-forms/get-started/installation?tabs=windows#android

Visual Studio 2017 Troubleshooting

1. **A number of errors and warnings are listed after you create a new solution, or yellow icons appear next to your solution's dependencies in the Solution Explorer:**

   - Make sure that you have updated the NuGet packages for the solution:

     1. Right-click on the solution (not on any of the projects).
     2. Select **Manage NuGet packages...**
     3. Click **Updates.**
     4. Select all the packages.
     5. Click **Update** to install the updates.

   - Clean the solution (right-click the solution > Clean Solution)

   - Build the solution (right-click the solution > Build Solution).

   - Close Visual Studio 2017 and re-open your project.

   The errors eventually seem to go away. If they still don't go away, you can also try one or more of the following:

   - Right-click on the solution and click *Restore NuGet Packages*.

   - Right-click your project (just below the solution), click *Properties*, take note of the *Target framework* setting, change the target framework to the previous one, build your project, change the target framework back to the correct/original setting, and build your project again.

2. **Error "Unexpected install output: Error: android.os.ParcelableException: java.io.IOException: Requested internal only, but not enough space" occurs when trying to deploy your app to the Android emulator.**

   - Restart the Android emulator by using the emulator's interface (click and hold the power button, then select restart).

   - If that does not help, do the following to factory reset the emulator device:

     1. Select Tools > Android >Android Device Manager…
     2. Find the emulator you are using in the list.
     3. Stop it if it is running by clicking the **Stop** button.
     4. Right-click on it and select **Factory Reset**.

3. **Error "Unable to access jarfile C:\Program Files (x86)\Android\android-sdk\build-tools\32.0.0\lib\dx.jar" occurs when trying to deploy your app to the Android emulator.**

   - Start Visual Studio 2017 as Administrator (Start > right-click *Visual Studio 2017* > More > Run as administrator).
   - Go to Tools > Android > Android SDK Manager.
   - If the Android SDK Location box is empty, click the … and browse to the *C:\Program Files (x86)\Android\android-sdk* folder

- If no SDK component information is listed, you may need to click the settings icon in the bottom-right corner of the window and select Repository > Google
- Click on the *Tools* tab
- Uncheck Android SDK Build-Tools 32 to uninstall it and apply the changes

4. The emulator flashes, is unresponsive or says that the phone data is corrupt when you try to interact with the phone.

- Start Visual Studio 2017 as Administrator (Start > right-click *Visual Studio 2017* > More > Run as administrator).
- Go to Tools > Android > Android Device Manager.
- Stop the emulator if it is running.
- Edit it and change a setting but change it back to what it was.
- Click the *Save* button.

5. Error "The type or namespace name 'App' could not be found" appears in the Output window when you try to build and run your app

- This seems to be caused when you the name that you chose for your app contains a hyphen or some other unusual character. It is best to name your app without using hyphens, spaces, etc (e.g. “MyApp” instead of “My App”). Alternatively, try the steps below.
- Right click on the Android project, click Add > Reference.
- Expand “Projects” on the left and click “Solution”.
- Tick the name that you gave your project (not the project with the .iOS or Android suffix) and click the *OK* button.